

When and Why SignSGD Outperforms SGD: A Theoretical Study Based on ℓ_1 -norm Lower Bounds

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1. TL;DR

A provable advantage for SignSGD. Under ℓ_∞ -smoothness and coordinate-wise noise, SignSGD has a better worst-case convergence rate than SGD when noise is sparse or highly heterogeneous.

SignSGD: upper bound

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \|\nabla f(\mathbf{x}_t)\|_1 \right] = \mathcal{O} \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right).$$

SGD: lower bound

$$\mathbb{E} \left[\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d L_\infty \Delta}{T}} + \left(\frac{d \|\sigma\|_1^2 L_\infty \Delta}{BT} \right)^{\frac{1}{4}} \right).$$

Up to d -fold fewer gradient samples under sparse noise.

$N = BT$: gradient samples; d : coordinates; Δ : initial objective gap.

2. Why change the setting?

Practice creates a puzzle.

SignSGD and Muon reach lower losses in language-model training. Yet SGD is minimax-optimal in the standard stochastic ℓ_2 framework. A provable advantage requires a problem class that exposes coordinate structure.

Sign updates fit ℓ_1 - ℓ_∞ geometry.

SignGD is steepest descent under an ℓ_∞ step constraint:
 $\text{sign}(g) \in \arg \max_{\|v\|_\infty \leq 1} \langle g, v \rangle.$

Change the assumptions and the target.

Write $D_f(x, y) = f(y) - f(x) - \langle \nabla f(x), y - x \rangle$ for the local linearization error.

Standard framework	Our framework
Smoothness ℓ_2 -smoothness $ D_f(x, y) \leq (L_2/2) \ y - x\ _2^2$	Smoothness ℓ_∞ -smoothness $ D_f(x, y) \leq (L_\infty/2) \ y - x\ _\infty^2$
Noise One aggregate bound $\mathbb{E} \ g - \nabla f\ _2^2 \leq \sigma^2$	Noise Coordinate-wise bounds $\mathbb{E} (g_i - \nabla_i f)^2 \leq \sigma_i^2$ for each coordinate
Stationarity $\ \nabla f\ _2$	Stationarity $\ \nabla f\ _1$

Coordinate-wise bounds retain the differences among noise scales $\sigma_1, \dots, \sigma_d$ that a single aggregate bound discards.

3. Lower bounds and key ideas

SignSGD: a tight characterization

$$\mathbb{E} \left[\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{L_\infty \Delta}{N}} + \left(\frac{\|\sigma\|_1^2 L_\infty \Delta}{N} \right)^{\frac{1}{4}} \right).$$

This matches the upper-rate scaling and characterizes SignSGD's worst-case complexity in our problem class.

Key idea. Start with a one-dimensional objective whose slope remains nonzero over the region visited by the algorithm. Unbiased two-point noise can make nonzero gradients rare, causing sign updates to stall. Distribute smoothness and initial gap over coordinates to recover the $\|\sigma\|_1$ dependence.

SGD: the separation benchmark

$$\mathbb{E} \left[\min_{0 \leq t < T} \|\nabla f(\mathbf{x}_t)\|_1 \right] = \Omega \left(\sqrt{\frac{d L_\infty \Delta}{T}} + \left(\frac{d \|\sigma\|_2^2 L_\infty \Delta}{BT} \right)^{\frac{1}{4}} \right).$$

Under the same geometry and stationarity target, SGD retains explicit dimension dependence and a stochastic term governed by $d \|\sigma\|_2^2$, rather than $\|\sigma\|_1^2$.

Key idea. Build a one-dimensional gradient grid that limits SGD's progress. Lift the construction across coordinates and choose curvature allocations that recover its deterministic and stochastic terms.

Muon: from vectors to matrices

$$\mathbb{E} \left[\min_t \|\nabla F(\mathbf{W}_t)\|_* \right] = \Omega \left(\sqrt{\frac{L_* \Delta}{N}} + \left(\frac{\|\Sigma\|_*^2 L_* \Delta}{N} \right)^{\frac{1}{4}} \right).$$

The tight vector rate extends to matrix parameters.

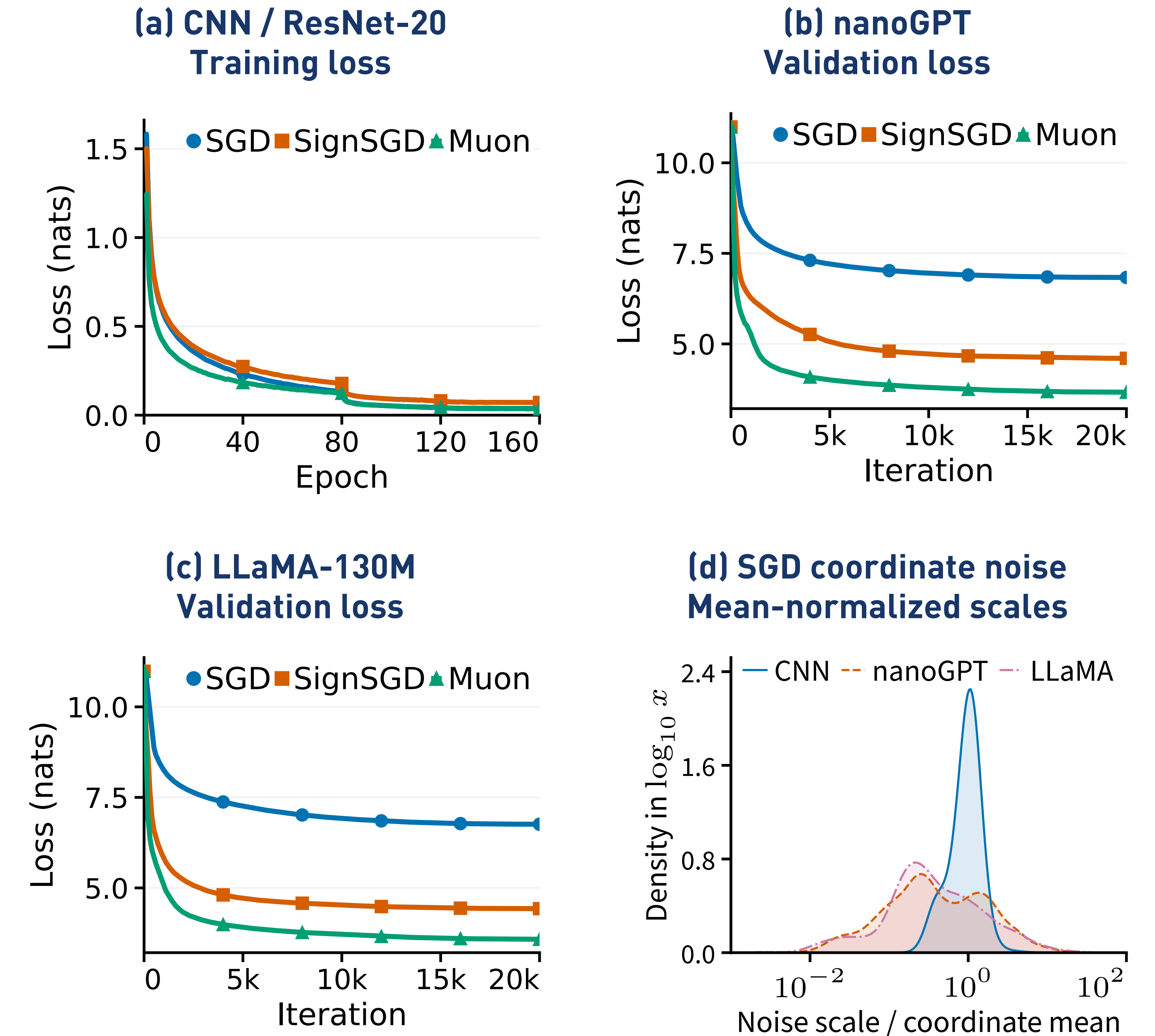
Object	Vector / SignSGD	Matrix / Muon
Step geometry	$\ \cdot\ _\infty$	$\ \cdot\ _{\text{op}}$
Stationarity	$\ \nabla f\ _1$	$\ \nabla F\ _*$
Noise scale	$\ \sigma\ _1$	$\ \Sigma\ _*$

Key idea. Embed the hard vector objective into a rotated diagonal matrix subspace. Exact matrix sign then reproduces the SignSGD trajectory. The matrix gradient's nuclear norm equals the vector gradient's ℓ_1 norm, so the hard instance and lower bound transfer.

Theory: exact, momentum-free matrix sign. Experiments: practical Muon.

Why upper vs. lower? A smaller upper bound alone may reflect sharper analysis. Comparing SignSGD's upper bound with SGD's lower bound establishes a genuine worst-case separation for this problem class.

4. Experiments



No SignSGD gain on CNN; clear gains on language models.

Performance panels (a)-(c): three-seed means.

Final C4 validation loss (nats)

Model	SGD	SignSGD	Muon
nanoGPT	6.835	4.603	3.666
LLaMA-130M	6.759	4.428	3.583

Noise heterogeneity connects to the theory.

From the lower bound compare we can see that SignSGD wins over SGD when $\chi > 1$, and the two optimizers are comparable when $\chi \approx 1$. We then estimated the coordinate-wise variance σ along SGD training, as presented in Figure (d):

CNN: similar performance, nearly uniform noise. The mean-normalized coordinate noise scales cluster near 1, and $\hat{\chi}$ is close to 1.

LLMs: clear SignSGD gains, heterogeneous noise. Their normalized scales are more dispersed across coordinates, and $\hat{\chi}$ is well above 1.

Our stochastic complexity terms are comparable when $\chi \approx 1$ and separate as χ grows.

The observed CNN-LLM contrast is consistent with this theoretical separation